

Meet Patel

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EDUCATION

University of Houston Clear Lake

MS in Computer Information System — GPA: 3.75

Houston, TX, USA

Jan. 2024 – Dec. 2025

Charutar Vidhya Mandal University

Bachelor of Computer Application — GPA: 9.98

Gujarat, India

Mar. 2020 – May 2023

TECHNICAL SKILLS

Programming & Analytics: Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), SQL, R

AI & ML Frameworks: LangGraph, LangChain, OpenAI GPT Models, Google Gemini, TensorFlow, PyTorch

Data Visualization: Power BI, Tableau, Excel (Pivot Tables, VBA, X-Lookup), Plotly

Databases & Cloud: MySQL, MongoDB, Snowflake, RDBMS

Development Tools: Git, Jupyter Notebook, Google Colab, REST APIs, TCP Sockets, Threading

ML Techniques: Regression, Classification, Clustering, Time-Series Forecasting, Gradient Descent, TDD

PROJECTS

Agentic AI for Covered-Call Option Screening | [Check out](#) | *Python, LangGraph, LangChain, OpenAI* **Aug.2025 - Present**

- **Led design and development** of an Agentic AI application **automating covered-call option screening** and recommendations for retail investors, serving as **Technical Lead** for backend development and data integration
- **Implemented Test-Driven Development (TDD)** and reproducible workflows to address non-deterministic AI behavior in financial decision systems, **ensuring reliable outputs across 100+ test scenarios**
- **Engineered multi-agent architecture** integrating **real-time market data** from Polygon.io and Alpha Vantage APIs, enabling **accurate analysis** of strike prices, premiums, delta values, and expiration parameters
- Built modular system with Options Data Agent for market retrieval and Strategy Agent for **AI-driven ranking** of contracts based on ROI and risk exposure, **improving decision-making efficiency by 40%**
- **Delivered interactive Streamlit web interface** for portfolio insights visualization, facilitating intuitive option recommendations and **multi-trade opportunity identification**

Custom Linear Regression for Housing Price Prediction | *Python, NumPy, Gradient Descent*

May 2025

- **Implemented from-scratch Linear Regression model** with Gradient Descent in Python to predict California housing prices, **achieving MAE of \$349** and demonstrating **99.5% accuracy** compared to scikit-learn baseline
- Curated and preprocessed California Housing dataset with **manual feature scaling**, enabling stable gradient updates and **30% faster convergence** during model training
- **Designed comprehensive training and evaluation workflow** reporting MAE and MSE metrics, establishing **clear quantifiable baselines** for targeted error analysis and model improvement

U.S. COVID-19 Vaccination Analysis | *Python, Pandas, Matplotlib, Seaborn, Plotly*

Mar. 2025 - Present

- **Led comprehensive analysis** of COVID-19 vaccination disparities across **3,000+ U.S. counties** using Python data science stack, **uncovering critical urban-rural and demographic trends** affecting vaccine distribution
- **Merged and cleaned large-scale datasets** from CDC and U.S. Census Bureau, standardizing FIPS codes and demographic data to **ensure 99% data integrity** across **10M+ records**
- **Visualized geospatial and time-series insights** via choropleth maps and interactive plots to **support data-driven public health decisions** and policy recommendations
- **Automated repeatable analyses** with modular Jupyter notebooks, **improving maintainability** and enabling reproducible research workflows

Secure Distributed Service Broker System | *Python, TCP Sockets, Threading, Tkinter*

Jan. 2025

- **Designed and implemented secure distributed system** featuring service broker and multiple providers for seamless service discovery, **handling 50+ concurrent connections** with **99% uptime**
- **Implemented multithreaded server architecture** using Python threading to manage concurrent client connections, **improving system responsiveness under load by 60%**
- **Secured service access** with ID/password authentication and **cryptographic hashing** (MD5, SHA-256, SHA-512), **enforcing credential verification** and strengthening data integrity
- **Developed user-friendly Tkinter-based GUI client**, enhancing user experience and **reducing interaction complexity by 45%**

RELEVANT COURSEWORK

Data Warehouse & Business Intelligence, Artificial Intelligence, Big Data Analytics, Machine Learning, Predictive Analysis, Advanced Operating Systems, Database Management Systems, Data Mining, Data Structures & Algorithms, Statistics & Probability